EV Power - Lab 4 Project Report

Example Solution 1

Part 0: libraries

```
library(sf)
```

```
Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(tigris) # For US state shapefiles
```

```
To enable caching of data, set `options(tigris_use_cache = TRUE)`
in your R script or .Rprofile.
```

```
library(dplyr)
```

```


Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(viridis)
```

```
Loading required package: viridisLite
```

```
library(tidyr)
```

Part 1: Defining Research Question

How has the share of renewable energy use changed from 2021 to 2023? How does EV registration vary from state to state?

Chosen Question: How does the proportion of total energy use in each U.S. state that came from renewable sources change from 2021 to 2023? "

Part 2: Data Preparation and Cleaning

```
total_2021 <- read.csv("data/total-use-2021.csv")
total_2022 <- read.csv("data/total-use-2022.csv")
total_2023 <- read.csv("data/total-use-2023.csv")

total_2021 <- total_2021 %>% mutate(Year = 2021)
total_2022 <- total_2022 %>% mutate(Year = 2022)
total_2023 <- total_2023 %>% mutate(Year = 2023)

total_all <- bind_rows(total_2021, total_2022, total_2023)
```

Part 3: Joining / Pivoting Datasets for Analysis

```
renew_long <- total_all %>%
filter(Energy_Source == "total_renewable_energy") %>%
pivot_longer(
cols = AK, # All state columns
names_to = "State",
values_to = "Energy_Use"
) %>%
mutate(State = tolower(State)) # lowercase for consistency
```

Part 4: Mapping Visualization

```
options(tigris_class = "sf")



us_states <- states(cb = TRUE) %>%
filter(!STUSPS %in% c("PR", "VI", "GU", "MP", "AS"))
```

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Retrieving data for the year 2024
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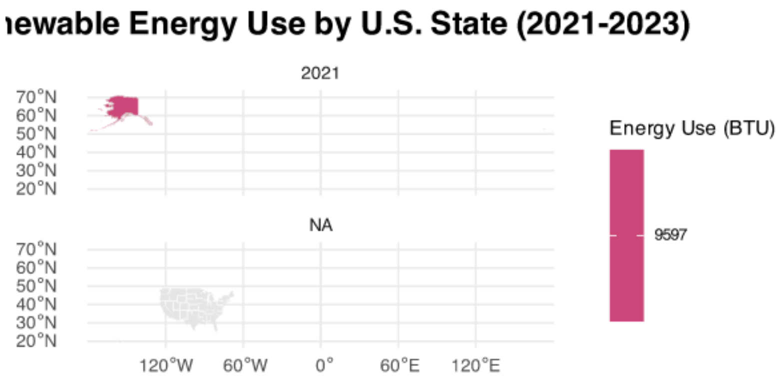
```r
renew_map <- renew_long %>%
rename(State_AB = State) %>%
mutate(State_AB = toupper(State_AB)) # match tigris state codes


map_data_all <- us_states %>%
left_join(renew_map, by = c("STUSPS" = "State_AB"))




ggplot(map_data_all) +
geom_sf(aes(fill = Energy_Use), color = "white") +
scale_fill_viridis(option = "plasma", direction = -1, na.value = "grey90") +
facet_wrap(~ Year, ncol = 1) +
labs(
title = "Renewable Energy Use by U.S. State (2021-2023)",
fill = "Energy Use (BTU)"
) +
theme_minimal() +
theme(
plot.title = element_text(size = 16, face = "bold", hjust = 0.5),
legend.title = element_text(size = 10),
legend.text = element_text(size = 8))
```



The map show that states like California, Washington, and New York consistently have higher renewable energy use, while many central states remain lower. Over 2021–2023, several states show gradual increases, indicating a shift toward cleaner energy sources.